Pentium Processor Architecture & Superscalar Operation MPCA Module 5

> **Definition:** The **Intel Pentium Processor** is a 32-bit superscalar CISC microprocessor capable of executing **two instructions per clock cycle** using two parallel 5-stage integer pipelines (**U-Pipe** and **V-Pipe**), an integrated on-chip Floating-Point Unit (FPU), and **Dynamic Branch Prediction Logic**.

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1. Detailed Technical Explanation

1. Pentium Architecture Block Diagram

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2. Superscalar Integer Pipelines (U and V Pipes)

The Pentium can issue two integer instructions simultaneously in parallel if they meet **Instruction Pairing Rules**:

1. **U-

The 5 Pipeline Stages:

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1. PF (
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3. Dynamic Branch Prediction & Branch Target Buffer (BTB)

- **Branch Target Buffer (BTB):** An on-chip cache storing branch instructions, target jump addresses, and branch history bits.

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4. Must-Write Points for Exams

- Pentium has a **64-bit external data bus** and **32-bit address bus**.

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5. Quick Recall Flow

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